

SHOYNA, Russian Federation

WMO: 222710

Lat: 67.877N

Lon: 44.174E

Elev: 26

StdP: 14.68

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-20.6	-15.7	-25.3	1.4	-20.5	-20.0	1.8	-15.4	40.6	15.4	34.7	13.4	4.4	90	0.900

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.2	69.5	60.6	65.2	58.7	61.7	56.6	62.3	67.4	59.4	63.8	57.1	61.1	12.2	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
59.8	76.9	65.2	57.2	69.9	62.4	54.9	64.1	59.5	28.1	67.7	26.1	63.7	24.5	61.2	72.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 31.4	27.6	24.6	-24.0	74.8	5.9	4.6	-28.2	78.1	-31.7	80.7	-35.0	83.3	-39.3	86.6		
(5)			-24.2	65.2	5.6	2.7	-28.3	67.1	-31.6	68.7	-34.7	70.3	-38.8	72.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	30.9	11.3	10.4	16.2	24.4	33.5	44.9	52.1	50.4	45.0	36.2	25.4	19.0		
	DBStd	16.95	12.73	13.91	10.61	7.77	6.94	7.17	6.91	4.98	4.90	6.07	9.25	11.20		
	HDD50	7228	1199	1109	1048	767	513	191	56	57	162	428	738	960		
	HDD65	12468	1664	1529	1513	1217	976	604	405	454	600	893	1188	1425		
	CDD50	241	0	0	0	0	2	38	121	68	12	0	0	0		
	CDD65	4	0	0	0	0	0	0	4	0	0	0	0	0		
	CDH74	30	0	0	0	0	0	1	28	1	0	0	0	0		
	CDH80	1	0	0	0	0	0	0	1	0	0	0	0	0		
	WSAvg	12.0	12.4	12.8	12.0	11.1	11.7	12.1	11.2	11.0	11.5	12.3	12.7	12.8		
	PrecAvg	16.5	1.1	0.9	1.0	0.9	1.2	1.3	1.4	1.9	1.9	2.2	1.4	1.3		
(15) Precipitation	PrecMax	23.3	3.0	1.6	2.8	2.1	2.8	2.5	3.3	4.3	3.8	3.9	3.0	2.6		
	PrecMin	13.1	0.4	0.2	0.3	0.2	0.4	0.4	0.4	0.6	0.5	1.1	0.4	0.4		
	PrecStd	2.0	0.5	0.4	0.6	0.5	0.6	0.6	0.7	0.9	0.8	0.7	0.6	0.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.4	33.5	34.2	37.5	60.7	70.8	78.0	69.7	60.1	49.3	42.0	35.8		
		MCWB	32.4	32.9	32.8	35.1	52.2	59.5	65.2	60.1	55.1	47.5	40.8	34.5		
	2%	DB	31.3	32.1	32.8	35.2	52.7	65.6	72.4	64.4	56.0	47.4	39.4	34.1		
		MCWB	30.3	31.5	31.8	33.6	47.4	57.2	63.2	59.6	53.1	45.8	37.9	32.9		
	5%	DB	29.9	30.0	31.5	34.1	47.1	60.9	67.8	61.1	54.0	45.7	37.4	32.8		
		MCWB	28.9	29.4	30.7	32.8	43.6	53.9	60.4	57.3	51.7	44.1	36.0	31.8		
	10%	DB	28.3	27.7	30.1	33.3	43.1	56.7	63.8	58.6	52.2	44.3	35.6	31.6		
		MCWB	27.4	27.0	29.3	32.2	40.6	51.4	58.3	55.4	50.1	42.7	34.2	30.7		
	0.4%	WB	32.6	33.2	33.2	35.4	52.6	61.9	67.4	62.8	56.5	47.9	40.8	34.9		
		MCDB	33.1	33.3	33.7	36.4	59.6	67.5	74.5	66.4	58.6	49.0	41.6	35.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	30.6	31.6	32.1	34.1	47.6	57.6	64.3	59.7	53.9	46.0	38.2	33.1		
		MCDB	31.1	31.9	32.6	34.8	52.5	64.0	71.7	63.1	55.6	47.1	39.1	33.8		
	5%	WB	29.1	29.4	30.8	33.2	43.8	54.7	61.5	57.6	52.0	44.3	36.3	32.0		
		MCDB	29.8	30.0	31.4	33.9	47.1	60.6	66.7	60.7	53.7	45.4	37.2	32.5		
	10%	WB	27.5	27.0	29.4	32.5	40.6	51.7	58.9	55.8	50.3	42.9	34.4	30.8		
		MCDB	28.3	27.6	30.1	33.3	43.0	56.4	63.2	58.4	52.0	44.1	35.3	31.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	12.3	11.6	11.9	9.9	8.4	10.3	11.2	9.0	7.3	5.9	8.1	10.3		
		MCDBR	10.2	7.6	7.7	7.2	16.1	18.5	18.3	14.1	10.2	5.6	5.7	6.4		
		MCWBR	10.0	7.8	7.6	6.7	11.6	11.7	10.9	9.7	7.5	5.3	5.4	6.2		
	5% WB	MCDBR	10.0	7.5	7.8	6.7	15.7	17.1	16.5	12.7	9.0	5.0	5.8	6.6		
		MCWBR	10.0	7.7	7.8	6.3	11.7	12.0	10.9	9.4	7.1	5.2	5.8	6.5		
(40) Clear-Sky Solar Irradiance	taub		0.219	0.264	0.302	0.330	0.339	0.339	0.370	0.348	0.323	0.292	0.363	N/A		
	taud		1.924	2.035	2.095	2.157	2.361	2.492	2.429	2.479	2.497	2.244	2.560	N/A		
	Ebn at Noon		27	163	227	257	271	275	259	254	231	156	34	0		
	Edh at Noon		3	16	28	35	32	29	30	25	19	13	4	0		
(44) All-Sky Solar Radiation	RadAvg		10	128	580	1128	1476	1652	1533	1012	541	168	24	0		
	RadStd		1	24	64	85	96	182	171	107	59	27	2	0		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page